

Internship Assignment

AI Agent for Automated Food Image Collection & Processing

Objective

Build an AI-powered automation agent that can take a list of food items from an Excel file, find a suitable image for each item online, process the image according to the required specifications, and automatically save the final images to a designated Google Drive folder.

The goal is to reduce the manual work involved in finding and preparing food images for a restaurant menu.

Input

You will be provided with an Excel file containing a list of food item names.

Food Item
Paneer Butter Masala
Dal Tadka
Veg Biryani
Butter Naan
Chicken Tikka

Required Workflow

1. Read the Excel file

- Read the food item names from the Excel sheet.
- Identify each item individually.
- Process multiple food items without requiring manual input for every item.

2. Search for the food image

- Search online for a suitable image for each food item.
- Possible sources include Google Images, other search engines, restaurant websites, Zomato restaurant listings, and other relevant food/restaurant websites.
- Select an image that accurately represents the food item.

3. Evaluate the image

- The dish should be clearly visible.
- The dish should be reasonably centred.
- The image should not be excessively zoomed in.
- The image should not be blurry or significantly cropped.
- The image should be suitable for use as a restaurant menu image.

4. Resize and process the image

- File format: JPG or PNG
- Dimensions: 1800 × 1200 pixels

- File size: below 10 MB
- Maintain good visual quality after resizing/compression.

5. Ensure the dish is properly positioned

- Food should be clearly visible and approximately in the centre of the frame.
- Food should not be excessively zoomed in.
- The dish should not be cropped unnecessarily.
- The image should be clear and not blurry.

6. Rename the image

- Rename the final image using the exact food item name from the Excel sheet.
- Example: Paneer Butter Masala.jpg, Dal Tadka.jpg, Veg Biryani.jpg.

7. Upload to Google Drive

- Automatically upload each processed image to a specified Google Drive folder.
- The final folder should contain the processed images with their corresponding food item names.

Important Requirement: Automation

The objective is not simply to manually find and resize images. The main objective is to build an automated agent/workflow that can perform as many of these steps as possible automatically:

Excel → Food Item → Image Search → Image Selection → Image Processing → Rename → Google Drive Upload

The fewer manual steps required, the better.

Bonus Features

- Automatically verify image quality by detecting blur, excessive zoom, poor framing, heavy cropping, or low resolution.
- Automatically search for another image if the first image does not meet the requirements.
- Intelligently handle difficult food names or variants such as “Special Paneer Tikka Masala Half.”
- Generate an Excel/CSV processing report showing Food Item, Image Found, Image Processed, Uploaded, Source, and Status.
- Use error handling so that a failed image search or upload does not stop the entire process.

Deliverables

- Working automation / AI agent
- Source code
- Instructions for running the agent
- Sample input Excel file
- Output Google Drive folder containing processed images
- Processing report showing the status of each food item

Evaluation Criteria

Criteria	Importance
Automation of the complete workflow	★★★★★
Ability to work with Excel	★★★★■
AI / agent implementation	★★★★★
Image search and selection	★★★★■
Image resizing / compression	★★★★■
Google Drive integration	★★★★■
Image quality validation	★★★██
Error handling	★★★██
Ease of running the system	★★★★■

Most important: We are looking for someone who can actually build the automation, not someone who simply knows how to use ChatGPT or other AI tools.

Suggested Technologies

Candidates are free to choose their own technology stack. They may use Python, APIs, ChatGPT, Claude, Gemini, LangChain, browser automation tools, image-processing libraries, or other suitable technologies, provided the final workflow works reliably.

Short Assignment Brief

Build an AI-powered automation that reads food item names from Excel, searches online for suitable images, selects an appropriate image, resizes it to 1800 × 1200 pixels, compresses it below 10 MB, ensures the dish is clear and centred, renames it using the food item's name, and uploads it to a specified Google Drive folder.

The complete workflow should ideally require minimal manual intervention: **Excel → Search → Select Image → Process → Rename → Google Drive.**